

2018 NYJC J2 H2 Mathematics Preliminary exam 9758/1 Marking Guide

Qn	Suggested Answers			Guidance
1	Let x , y and z be the no. of jeans, blouses and pairs of shoes Betty bought. $46.90 \times 0.85x + 29 \times 0.85y + 19.9z = 257.93$ $x + y + z = 10$ $2x - y = 0$ Using GC, $x = 2, y = 4, z = 4$ Hence, Betty bought 2 pairs of jeans, 4 blouses and 4 pairs of shoes.			

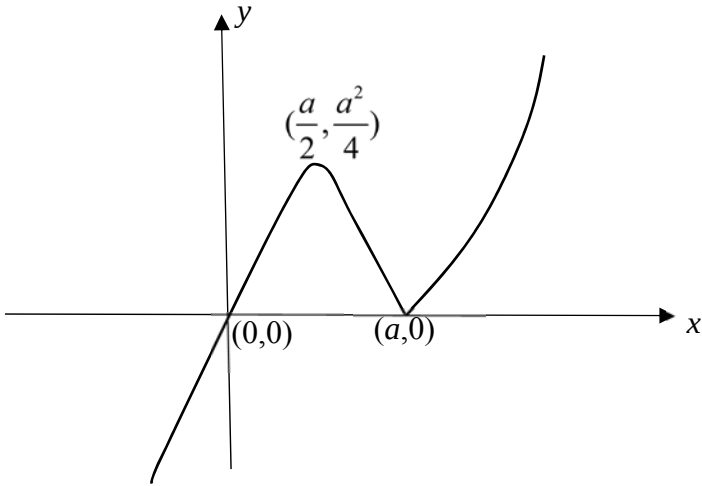
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2	$x^3 + y^3 = 3pxy$ Implicit differentiation w.r.t. x : $3x^2 + 3y^2 \frac{dy}{dx} = 3px \frac{dy}{dx} + 3py$ Gradient of asymptote = -1 Let $\frac{dy}{dx} = -1$. $3x^2 - 3y^2 = -3px + 3py$ $x^2 - y^2 = -px + py$ $(x+y)(x-y) = -p(x-y)$ $(x-y)(x+y+p) = 0$ $x = y$ or $x + y = -p$ (rejected since $p > 0$ and $x > 0, y > 0$) Since $x = y$, $x^3 + x^3 = 3px^2$ $x^2(2x - 3p) = 0$ $x = 0$ (rejected) or $x = \frac{3p}{2}$ Hence, P has coordinates $\left(\frac{3p}{2}, \frac{3p}{2}\right)$.			

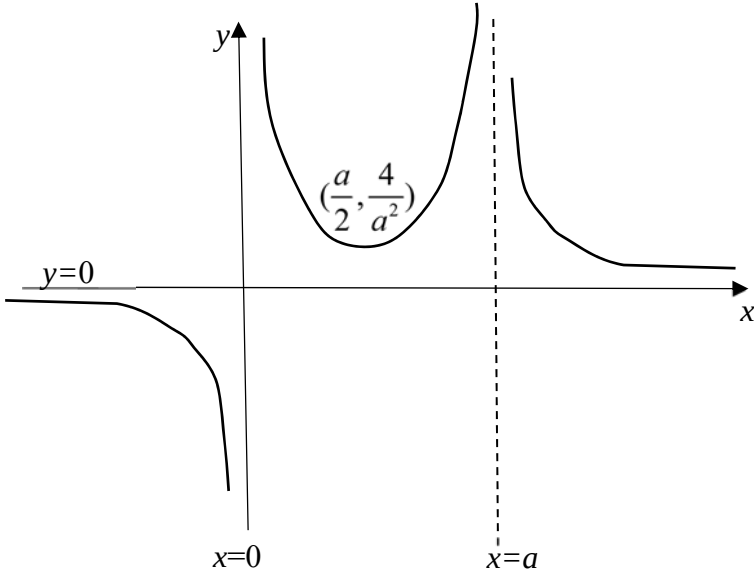
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3(i)	Let $x = k$ $y^2 + 14y + 4k^2 + 16k + 16ky + 13 = 0$ $y^2 + y(14 + 16k) + (4k^2 + 16k + 13) = 0$ For 2 distinct values of y , $(14 + 16k)^2 - 4(4k^2 + 16k + 13) > 0$ $196 + 448k + 256k^2 - 16k^2 - 64k - 52 > 0$ $240k^2 + 384k + 144 > 0$ Since $x = k$, $5x^2 + 8x + 3 > 0$ $(5x + 3)(x + 1) > 0$ $x < -1$ or $x > -\frac{3}{5}$ Therefore the set of x is $\{x : x \in \mathbb{R}, x < -1 \text{ or } x > -\frac{3}{5}\}$			
3(ii)	When $x = 0$ $y^2 + 14y + 13 = 0$ $(y + 13)(y + 1) = 0 \Rightarrow y = -1 \text{ or } -13$ The coordinates are $(0, -1)$ and $(0, -13)$ Curve C is not graph of a function as the vertical line $x = 0$ cuts the curve at 2 points.			

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4(i)	Reflection in the y-axis (Replace x with $-x$) $y = -x -x-a $ $= -x x+a $ Translation of 2 units in the direction of y-axis $y = 2 - x x+a $.			
4(ii)				

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4(iii)				
4(iv)	$0 < k < \frac{4}{a^2} \text{ or } k < 0$			

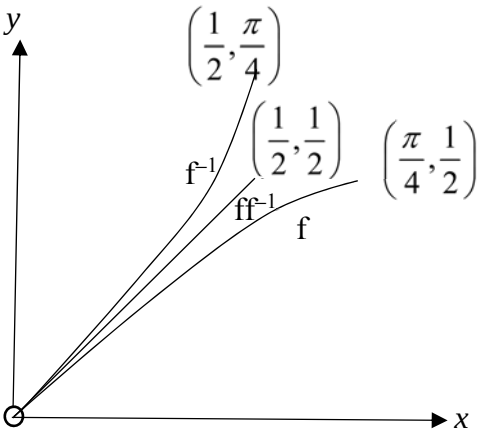
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5(i)	$l: \mathbf{r} = \begin{pmatrix} 5 \\ 2 \\ 4 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}, \beta \in \mathbb{R}$ Line perpendicular to plane passing through (3, 1, 1) is $\mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + \alpha \begin{pmatrix} 4 \\ 7 \\ 5 \end{pmatrix}$ The 2 lines will intersect at C. $\begin{pmatrix} 5+\beta \\ 2+3\beta \\ 4+\beta \end{pmatrix} = \begin{pmatrix} 3+4\alpha \\ 1+7\alpha \\ 1+5\alpha \end{pmatrix}$ $\beta - 4\alpha = -2$ $3\beta - 7\alpha = -1$ $\beta = 2, \alpha = 1$ Point C has coordinates (7, 8, 6)			
5(ii)	Since l does not intersect p_1 $\begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -2 \\ \lambda \end{pmatrix} = 0$ $3 - 6 + \lambda = 0 \Rightarrow \lambda = 3$			

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	$\begin{pmatrix} 5 \\ 2 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix} \neq \mu \quad \text{or} \quad \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix} \neq \mu$ $\mu \neq 23$			
5(iii)	Using $\lambda = 3$ $\frac{\left \begin{pmatrix} 5 \\ 2 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix} - \mu \right }{\left \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix} \right } = 2$ $ 23 - \mu = 2\sqrt{22}$ $23 - \mu = 2\sqrt{22} \quad \text{or} \quad 23 - \mu = -2\sqrt{22}$ $\mu = 23 + 2\sqrt{22} \quad \text{or} \quad \mu = 23 - 2\sqrt{22}$			

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6(i)	Since $g(-x) = \frac{1}{(-x)} = -\frac{1}{x} = -g(x)$, therefore g is odd. Although $f(-x) = \sin(-x) \cos(-x) = -\sin x \cos x = -f(x)$, but $f\left(-\frac{\pi}{2}\right) = -f\left(\frac{\pi}{2}\right) = 0$ and $f\left(-\frac{\pi}{2}\right)$ is undefined. Therefore, f is not odd.			
6(ii)	Since $f(0) = f\left(\frac{\pi}{2}\right) = 0$, there are two values of x that give the same value of $f(x)$, which means f is not one-one. OR: The line $y = 0$ cuts the graph $y = f(x)$ at 2 points. Therefore f has no inverse. $a = \frac{\pi}{4}$			
6(iii)				

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6(iv)	Domain of $f = \left(0, \frac{\pi}{4}\right]$ and Domain of $g = [-1, 0) \cup (0, 1]$			
	Range of $f = \left(0, \frac{1}{2}\right]$ and range of $g = (-\infty, -1] \cup [1, \infty)$			
	Since range of g is not a subset of domain of f , fg does not exist. Range of f is a subset of domain of g , therefore gf exists.			
	$gf(x) = g(\sin x \cos x) = \frac{1}{\sin x \cos x} = 2\operatorname{cosec} 2x$			
	Domain of gf is $\left(0, \frac{\pi}{4}\right]$ and range of gf is $[2, \infty)$.			

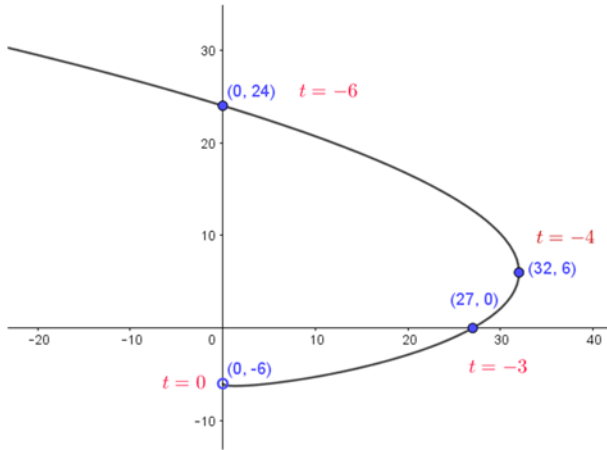
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Qn	Suggested Answers			Guidance
7(a)(i)	Let the angle between $\mathbf{a} + \mathbf{b}$ and $\mathbf{a}$ be α $(\mathbf{a} + \mathbf{b}) \cdot \mathbf{a} = \mathbf{a} + \mathbf{b} \mathbf{a} \cos \alpha$			
	$\cos \alpha = \frac{(\mathbf{a} ^2 + \mathbf{b} \cdot \mathbf{a})}{ \mathbf{a} + \mathbf{b} \mathbf{a} } = \frac{(1 + \mathbf{b} \cdot \mathbf{a})}{ \mathbf{a} + \mathbf{b} \mathbf{a} }$			
	Let the angle between $\mathbf{a} + \mathbf{b}$ and $\mathbf{b}$ be β $(\mathbf{a} + \mathbf{b}) \cdot \mathbf{b} = \mathbf{a} + \mathbf{b} \mathbf{b} \cos \beta$ $\cos \beta = \frac{(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} ^2)}{ \mathbf{a} + \mathbf{b} \mathbf{b} } = \frac{(\mathbf{a} \cdot \mathbf{b} + 1)}{ \mathbf{a} + \mathbf{b} \mathbf{b} }$ Since $\cos \alpha = \cos \beta$, $\mathbf{a} + \mathbf{b}$ is the angle bisector of AOB			
7(a)(ii)	$ \mathbf{a} \times \mathbf{b} = \mathbf{a} \mathbf{b} \sin AOB = \sin AOB$ Since area of triangle $= \frac{1}{2} \mathbf{a} \times \mathbf{b} = \frac{1}{\sqrt{10}}$ $\therefore \sin AOB = \frac{2}{\sqrt{10}}$			
7(b)(i)	$\vec{CD} = \vec{OD} - \vec{OC}$ $= \begin{pmatrix} 4 \\ 7 \\ 4 \end{pmatrix} - \begin{pmatrix} 7 \\ -8 \\ 7 \end{pmatrix} = \begin{pmatrix} -3 \\ 15 \\ -3 \end{pmatrix}$			

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	$\text{Perpendicular distance of } O \text{ from line} = \frac{\begin{vmatrix} -3 & 4 \\ 15 & 7 \\ -3 & 4 \end{vmatrix}}{\begin{vmatrix} -3 \\ 15 \\ -3 \end{vmatrix}}$ $= \frac{\begin{vmatrix} 81 \\ 0 \\ -81 \end{vmatrix}}{\sqrt{9+225+9}}$ $= \frac{81\sqrt{2}}{9\sqrt{3}} = 3\sqrt{6}$			
7(b)(ii)	$\vec{CO} \cdot \vec{CD} = \vec{CO} \vec{CD} \cos \theta$ $\begin{pmatrix} -7 \\ 8 \\ -7 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 15 \\ -3 \end{pmatrix} = \sqrt{7^2 + 8^2 + 7^2} \sqrt{3^2 + 15^2 + 3^2} \cos \theta$ $21 + 120 + 21 = \sqrt{162} \sqrt{243} \cos \theta$ $\theta = 35.3^\circ \text{ or } 0.615 \text{ radians}$ Alternatively, $\vec{OC} = \sqrt{49 + 64 + 49} = \sqrt{162}$ Let F be foot of perpendicular from O to line CD. Then $OF = 3\sqrt{6}$ $\theta = \sin^{-1} \left(\frac{3\sqrt{6}}{\sqrt{162}} \right) = 35.3^\circ$			

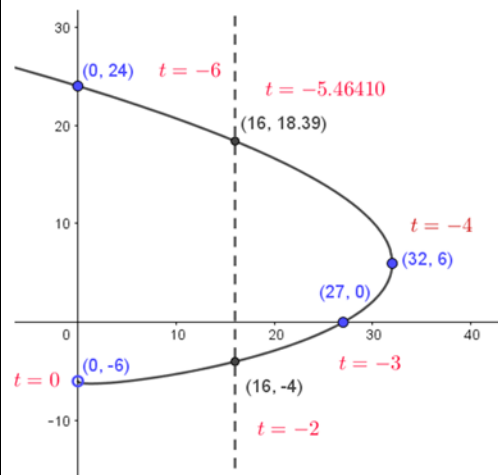
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8(i)	$\frac{dy}{dx} = \frac{dy}{dt} / \frac{dx}{dt} = \frac{2t+1}{3t^2+12t}$			
	For tangent parallel to the y -axis, $3t(t+4) = 0$			
	$t = -4, 0 \text{ (reject } t = 0 \text{)}$ Equation of tangent is $x = 32$			
8(ii)	When $x = 0$, $t = 0, -6$ corresponding to $(0, -6), (0, 24)$ When $y = 0$, $t = 2, -3$ (reject $t = 2$) corresponding to $(27, 0)$			
				
8(iii)	When $x = 16$			
	$t^2(t+6) = 16$			

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$t = -5.46410, -2, 1.46410$ (reject $t = 1.46410$ since $t < 0$)

The line $x = 16$ cuts C at $(16, -4)$ and $(16.0, 18.3923)$



$$\begin{aligned} \text{Required area} &= \int_{-2}^{-5.46410} (t^3 + 6t^2)(2t+1) dt - 16(18.39 - (-4)) \\ &= 241.058 ; 241 \text{ (3 s.f.)} \end{aligned}$$

Alternatively,

Required area

$$\begin{aligned} &= \int_{-5.46410}^{-4} (t^2 + t - 6)(3t^2 + 12t) dt - \int_{-2}^{-4} (t^2 + t - 6)(3t^2 + 12t) dt \\ &= 241.02 ; 241 \text{ (3 s.f.)} \end{aligned}$$

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9(a)(i)	$S_{\infty} = \frac{6}{13}$ $ S_n - S_{\infty} < 10^{-8}$ $\left \frac{6}{13} \times \frac{1}{3^{3n}} \right < 10^{-8}$ Using GC, smallest $n = 6$			
9(a)(ii)	$u_n = S_n - S_{n-1}$ $= \frac{6}{13} \left(1 - \frac{1}{3^{3n}} \right) - \frac{6}{13} \left(1 - \frac{1}{3^{3n-3}} \right)$ $= \frac{6}{13} \left(\frac{1}{3^{3n-3}} - \frac{1}{3^{3n}} \right)$ $= \frac{6}{13} \times \frac{1}{3^{3n}} \left(\frac{1}{3^{-3}} - 1 \right)$ $= \frac{6}{13} \times \frac{1}{3^{3n}} (26)$ $= \frac{12}{3^{3n}} = \frac{4}{3^{3n-1}}$			
9(b)(i)	$\sin(2r+1)\theta - \sin(2r-1)\theta$ $= \sin 2r\theta \cos \theta + \cos 2r\theta \sin \theta - \sin 2r\theta \cos \theta + \cos 2r\theta \sin \theta$ $= 2 \cos 2r\theta \sin \theta \text{ (shown)}$			
9(b)(ii)	$\sum_{r=1}^n \sin^2 r\theta$			

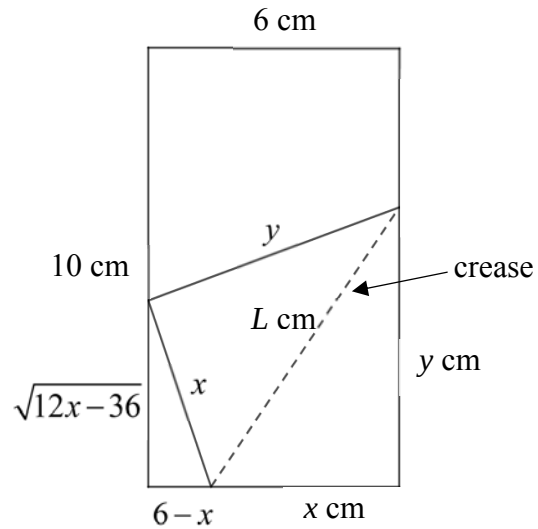
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	$= \frac{1}{2} \sum_{r=1}^n (1 - \cos 2r\theta)$			
	$= \frac{1}{2} \sum_{r=1}^n 1 - \frac{1}{2} \sum_{r=1}^n \cos 2r\theta$			
	$= \frac{1}{2} n - \frac{1}{4 \sin \theta} \sum_{r=1}^n [\sin(2r+1)\theta - \sin(2r-1)\theta], \theta \neq k\pi$			
	$= \frac{1}{2} n - \frac{1}{4 \sin \theta} [\sin 3\theta - \sin \theta$ $\quad + \sin 5\theta - \sin 3\theta$ $\quad + \sin 7\theta - \sin 5\theta$ $\quad \cdot$ $\quad \cdot$ $\quad + \sin(2n+1)\theta - \sin(2n-1)\theta]$			
	$= \frac{1}{2} n - \frac{1}{4 \sin \theta} [\sin(2n+1)\theta - \sin \theta]$			
	$= \frac{1}{2} n - \frac{1}{4 \sin \theta} [2 \cos(n+1)\theta \sin n\theta]$			
	$= \frac{1}{2} n - \frac{1}{2 \sin \theta} [\cos(n+1)\theta \sin n\theta]$			

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10(i)	$\frac{m}{\sin(\pi - (\theta + \alpha))} = \frac{l}{\sin \alpha}$ $l = \frac{m \sin \alpha}{\sin(\pi - (\theta + \alpha))}$ $= \frac{m \sin \alpha}{\sin(\theta + \alpha)} \text{ (shown)}$			
10(ii)	$l = \frac{m \sin \alpha}{\sin(\theta + \alpha)}$ $\frac{dl}{d\theta} = \frac{-(m \sin \alpha) \cos(\theta + \alpha)}{\sin^2(\theta + \alpha)}$ Alternatively, $\frac{dl}{d\theta} = -(m \sin \alpha) \operatorname{cosec}(\theta + \alpha) \cot(\theta + \alpha)$			
10(iii)	Using chain rule, $\frac{dl}{dt} = \frac{dl}{d\theta} \frac{d\theta}{dt}$ $= \frac{-15 \left(\sin \frac{\pi}{60} \right) \cos \left(\frac{\pi}{9} + \frac{\pi}{60} \right)}{\sin^2 \left(\frac{\pi}{9} + \frac{\pi}{60} \right)} \left(-\frac{5\pi}{9} \right)$ $= 8.2611 \text{ cm / sec}$ $\approx 8.26 \text{ cm / sec}$			
10(iv)	$L = \sqrt{x^2 + y^2}$ $\sqrt{x^2 - (6 - x)^2} = \sqrt{x^2 - 36 + 12x - x^2} = \sqrt{12x - 36}$			

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$$\text{Area of rectangle} = 6 \times 10 = 60 \text{ cm}^2$$

$$60 = \frac{1}{2}(6-x)(\sqrt{12x-36}) + 2\left(\frac{1}{2}\right)(xy)$$

$$+ \frac{1}{2}(6)(10 - \sqrt{12x-36} + 10 - y)$$

$$= \frac{1}{2}(6-x)(\sqrt{12x-36}) + xy + 60 - 3\sqrt{12x-36} - 3y$$

$$0 = \frac{1}{2}(6-x)(\sqrt{12x-36}) + xy - 3\sqrt{12x-36} - 3y$$

$$3y - xy = \left(3 - \frac{x}{2}\right)(\sqrt{12x-36}) - 3\sqrt{12x-36}$$

$$y = \frac{-\frac{x}{2}(\sqrt{12x-36})}{3-x}$$

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	$= \frac{x(\sqrt{12x-36})}{2x-6}$ $L = \sqrt{x^2 + y^2}$ $= \sqrt{x^2 + \left(\frac{x(\sqrt{12x-36})}{2x-6}\right)^2}$ $= \sqrt{x^2 + \left(\frac{x^2(12x-36)}{(2x-6)^2}\right)}$ $= \sqrt{x^2 + \left(\frac{12x^2(x-3)}{4(x-3)^2}\right)} = \sqrt{x^2 + \frac{3x^2}{x-3}}$			
10(v)	$\frac{dL}{dx} = \frac{1}{2} \left(x^2 + \frac{3x^2}{x-3}\right)^{-\frac{1}{2}} \left(2x + \frac{(x-3)(6x) - (3x^2)}{(x-3)^2}\right)$ $= \frac{2x + \frac{3x^2 - 18x}{(x-3)^2}}{2\sqrt{x^2 + \frac{3x^2}{x-3}}} = 0$ $2x + \frac{3x^2 - 18x}{(x-3)^2} = 0$ $x \left(2 + \frac{3x-18}{(x-3)^2}\right) = 0$			

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$$2 + \frac{3x-18}{(x-3)^2} = 0 \text{ or } x = 0 \text{ (rejected)}$$

$$2 + \frac{3x-18}{(x-3)^2} = 0$$

$$\frac{2(x-3)^2 + 3x - 18}{(x-3)^2} = 0$$

$$2(x^2 - 6x + 9) + 3x - 18 = 0$$

$$2x^2 - 9x = 0$$

$$x = \frac{9}{2} \text{ or } x = 0 \text{ (rejected)}$$

By first derivative test,

x	4.45	$\frac{9}{2}$	4.55
$\frac{dL}{dx}$	-0.0604	0	0.0553
Slope	\	—	/

$$x = \frac{9}{2} \text{ gives a minimum } L.$$

Hence,

$$\text{Min. } L = \sqrt{\left(\frac{9}{2}\right)^2 + \frac{3\left(\frac{9}{2}\right)^2}{\left(\frac{9}{2}\right) - 3}} = 7.79\text{cm}$$

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11(i)	If the rate of change of x is not positive, it is not possible for the amount of drug to increase to reach 120 mg in the body for the treatment to be effective.			
11(ii)	$\frac{dx}{dt} = 30 - kx, \text{ where } k > 0$ $\Rightarrow \int \frac{1}{30 - kx} dx = \int dt$ $\Rightarrow -\frac{1}{k} \ln(30 - kx) = t + C, \text{ since } \frac{dx}{dt} = 30 - kx > 0$ $\Rightarrow \ln(30 - kx) = -kt - kC$ $\Rightarrow (30 - kx) = e^{-kt - kC} = e^{-kC} e^{-kt} = B e^{-kt}, B > 0$ When $x = 0, t = 0$, thus $B = 30$ When $t = 4, x = 82.6$, $30 - k(82.6) = 30e^{-4k}$, Solving using GC. $k = 0.200$ $x = 150 \left(1 - e^{-\frac{t}{5}} \right), \text{ where } A = 150$			
11(iii)	When $x = 120, k = \frac{1}{5}$, $120 = 150 \left(1 - e^{-\frac{t}{5}} \right)$ $\Rightarrow t = 8.05 \text{ hrs}$			
11(iv)	$\frac{d^2x}{dt^2} = \frac{1}{\sqrt{2500 - 9t^2}}$			

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$\frac{dx}{dt} = \int \frac{1}{\sqrt{2500-9t^2}} dt = \frac{1}{3} \sin^{-1}\left(\frac{3t}{50}\right) + C$ $x = \int \left[\frac{1}{3} \sin^{-1}\left(\frac{3t}{50}\right) + C \right] dt$ $= \frac{1}{3} \left[t \sin^{-1}\left(\frac{3t}{50}\right) - \int t \cdot \frac{1}{\sqrt{1-\left(\frac{3t}{50}\right)^2}} \left(\frac{3}{50}\right) dt \right] + Ct + D$ $= \frac{1}{3} \left[t \sin^{-1}\left(\frac{3t}{50}\right) - \int t \cdot \frac{3}{\sqrt{50^2-(3t)^2}} dt \right] + Ct + D$ $= \frac{1}{3} \left[t \sin^{-1}\left(\frac{3t}{50}\right) + \frac{1}{6} \int \frac{-18t}{\sqrt{2500-9t^2}} dt \right] + Ct + D$ $= \frac{1}{3} \left[t \sin^{-1}\left(\frac{3t}{50}\right) + \frac{1}{3} \sqrt{2500-9t^2} \right] + Ct + D$ $= \frac{1}{3} t \sin^{-1}\left(\frac{3t}{50}\right) + \frac{1}{9} \sqrt{2500-9t^2} + Ct + D$ When $t = 0$, $x = 0$, $D = -\frac{50}{9}$ $x = \frac{1}{3} t \sin^{-1}\left(\frac{3t}{50}\right) + \frac{1}{9} \sqrt{2500-9t^2} + Ct - \frac{50}{9}$			